Elliptic Surfaces and the Picard-Fuchs Equation

by Ryan Lam

The discussion below follows the note posted by Schnell.

Introduction

Our main object of consideration is elliptic surface:

> **Definition.** An *elliptic curve* E is a level set $E := \{(x, y, z) : F(x, y, z) = 0\}$ of a degree 3 homogenous complex polynomial $F \in \mathbb{C}[x, y, z]$.

Note that F can be "de-homogenize" to a function in two variable,

$$f(x, y) := F(x, y, 1)$$

> **Theorem.** Any elliptic curve E has form of $y^2 = x^3 + ax + b$ for some $a, b \in \mathbb{C}$ with $4a^23 + 27b^2 \neq 0$. This is called the *normal form* of E.

Proof is skipped or found in any ordinary text on elliptic curves.

> **Definition.** An *elliptic surface* is a family of elliptic curve E_t controlled by a parameter $t \in \mathbb{C}$ (?). That is, it is a level set of a family of homogenous degree 3 polynmoial $F_t(x, y, z)$.

Hence we can "de-homogenize" F_t and write E_t into normal form, namely, let:

$$y^2 = x^3 + a(t)x + b(t)$$

be the elliptic curve E_t, where $a(t), b(t) \in \mathbb{C}(t)$.

> From here on, all the variables depends on t, and we will suppress it automatically.

There is a holomorphic 1-form ω (on where?) we are interested in (for now I will just take it):

$$\omega = \frac{dx}{y} = \frac{dx}{\sqrt{x^3 + a(t)x + b(t)}},$$

for which we will write $\omega = g(x, t)\, dx$. Note that we can differentiate ω with respect to t as

$$\frac{d\omega}{dt} := \frac{\partial g(x, t)}{\partial t} dx.$$

And hence $\dfrac{d\omega}{dt}$ is still a 1-form.

> **Definition.** The *Picard-Fuchs* equation associated to E_t, is the second-order homogeneous differential equation where $\pi = \displaystyle\int_\gamma \omega$, where γ is any 1-chain. It has the form of
>
> $$\frac{d^2\pi}{dt^2} + B\frac{d\pi}{dt} + C\pi = 0$$
>
> where B, C are functions of t.

In this document we go through the computation of the B and C for various examples.

Example 1

Following Schnell, we consider the following elliptic surface:

$$F := t(x^3 + y^3 + z^3) - 3xyz$$

written in normal form, we have

$$E : y^2 = tx^3 + \frac{1}{4}\left(1 - 4t^3\right)x^2 + \frac{1}{3}t^2\left(t^3 - 1\right)x - \frac{1}{27}t\left(t^3 - 1\right)^2$$

Hence the 1-form ω is:

$$\omega = \frac{dx}{y} = \frac{1}{\sqrt{tx^3 + \frac{1}{4}\left(1 - 4t^3\right)x^2 + \frac{1}{3}t^2\left(t^3 - 1\right)x - \frac{1}{27}t\left(t^3 - 1\right)^2}}\,dx$$

For notational convienience, we write

$$y = \sqrt{tx^3 + \frac{1}{4}\left(1 - 4t^3\right)x^2 + \frac{1}{3}t^2\left(t^3 - 1\right)x - \frac{1}{27}t\left(t^3 - 1\right)^2}$$

The first step is to compute the one form

$$\eta = \frac{d^2\omega}{dt^2} + B\frac{d\omega}{dt} + C\omega$$

such that $\eta = \dfrac{P(x)}{y^5}$ for some polynomial $P(x)$ where x has 6 degrees.

In [1]:
```
x, t, B, C =var('x t B C')
y = sqrt(t*(x^3)+(1/4)*(1-4*t^3)*(x^2)+(1/3)* (t^2)*(t^3-1)*x-(1/27)*t*(t^3-1)^2)
f = 1/y
f_dot = f.diff(t)
f_ddot = f_dot.diff(t)
eta = f_ddot+B*f_dot+C*f
eta = eta.factor()
show(eta)
```

$$\frac{6\sqrt{3}\left(16\,Ct^{14} - 56\,Bt^{13} - 288\,Ct^{12}x + 864\,Bt^{11}x + 2160\,Ct^{10}x^2 - 64\,Ct^{11} + 252\,t^{12} - 5400\,Bt^9x^2 - 8640\,Ct^8x^3 + 176\,Bt^{10} + 864\,Ct^9x - 3096\,t^{10}x + 17280\,Bt^7x^3 + 19440\,Ct^6x^4 - 1944\,Bt^8x - 4536\,Ct^7x^2 + 14580\,t^8x^2 - 29160\,Bt^5x^4 - 23328\,Ct^4x^5 + 96\,Ct^8 - 480\,t^9 + 7938\,Bt^6x^2 + 11448\,Ct^5x^3 - 32400\,t^6x^3 + 23328\,Bt^3x^5 + 11664\,Ct^2x^6 - 192\,Bt^7 - 864\,Ct^6x + 3888\,t^7x - 14310\,Bt^4x^3 - 13608\,Ct^3x^4 + 34020\,t^4x^4 - 5832\,Btx^6 + 1296\,Bt^5x + 2592\,Ct^4x^2 - 10692\,t^5x^2 + 10206\,Bt^2x^4 + 5832\,Ctx^5 - 17496\,t^2x^5 - 64\,Ct^5 + 216\,t^6 - 2592\,Bt^3x^2 - 2808\,Ct^2x^3 + 9720\,t^3x^3 - 1458\,Bx^5 + 8748\,x^6 + 80\,Bt^4 + 288\,Ct^3x - 1080\,t^4x + 1404\,Btx^3 + 729\,Cx^4 + 972\,tx^4 - 216\,Bt^2x - 216\,Ctx^2 + 1944\,t^2x^2 + 16\,Ct^2 + 54\,Bx^2 + 324\,x^3 - 8\,Bt + 288\,tx + 12\right)}{\left(-4\,t^7 + 8\,t^4 + 108\,tx^3 - 27\left(4\,t^3 - 1\right)x^2 + 36\left(t^5 - t^2\right)x - 4\,t\right)^{\frac{5}{2}}}$$

In [2]:
```
# Extract the polynomial
eta_poly = eta.operands()[1]
show(QQ[B, C, t][x](eta_poly))
```

$$\left(11664Ct^2 - 5832Bt + 8748\right)x^6 + \left(-23328Ct^4 + 23328Bt^3 + 5832Ct - 17496t^2 - 1458B\right)x^5$$
$$+ \left(19440Ct^6 - 29160Bt^5 - 13608Ct^3 + 34020t^4 + 10206Bt^2 + 729C + 972t\right)x^4$$
$$+ \left(-8640Ct^8 + 17280Bt^7 + 11448Ct^5 - 32400t^6 - 14310Bt^4 - 2808Ct^2 + 9720t^3 + 1404Bt + 324\right)x^3$$
$$+ \left(2160Ct^{10} - 5400Bt^9 - 4536Ct^7 + 14580t^8 + 7938Bt^6 + 2592Ct^4 - 10692t^5 - 2592Bt^3 - 216Ct + 1944t^2 + 54B\right)x^2$$
$$+ \left(-288Ct^{12} + 864Bt^{11} + 864Ct^9 - 3096t^{10} - 1944Bt^8 - 864Ct^6 + 3888t^7 + 1296Bt^5 + 288Ct^3 - 1080t^4 - 216Bt^2 + 288t\right)x + 16Ct^{14} - 56Bt^{13} - 64Ct^{11} + 252t^{12}$$
$$+ 176Bt^{10} + 96Ct^8 - 480t^9 - 192Bt^7 - 64Ct^5 + 216t^6 + 80Bt^4 + 16Ct^2 - 8Bt + 12$$

Note that indeed η has a six degree polynomial at the denominator, and the numerator is a multiple of y^5.

The next step is to find constants $A_0, A_1, \dots A_4$ (depending on t) such that

$$\eta + A_4 d\left(\frac{x^4}{y}\right) + A_3 d\left(\frac{x^3}{y}\right) + A_2 d\left(\frac{x^2}{y}\right) + A_1 d\left(\frac{x}{y}\right) + A_0 d\left(\frac{1}{y}\right) = \frac{Q(x)}{y^5}$$

.

where $Q(x)$ is a polynomial of degree 1.

Note that $\frac{x^k}{y}$ is a function $h(x,t)$, so the d operator is just bring it to a one form:

$$d(h(x,t)) = \frac{\partial h(x,t)}{\partial x}dx.$$

In [3]:
```
A_0, A_1, A_2, A_3, A_4 = var("A_0 A_1 A_2 A_3 A_4")
```

In [4]:
```
f_0 = A_0*(f^3)
f_1 = A_1*x/(y^3)
f_2 = A_2*x^2/(y^3)
f_3 = A_3*x^3/(y^3)
f_4 = A_4*x^4/(y^3)
d_0 = f_0.diff(x)
d_1 = f_1.diff(x)
d_2 = f_2.diff(x)
d_3 = f_3.diff(x)
d_4 = f_4.diff(x)
find_As_exp = eta+d_0+d_1+d_2+d_3+d_4
find_As_exp = find_As_exp.factor()
show(find_As_exp)
```

$$\frac{6\sqrt{3}\left(16\,Ct^{14} - 56\,Bt^{13} - 288\,Ct^{12}x + 864\,Bt^{11}x + 2160\,Ct^{10}x^2 - 64\,Ct^{11} + 252\,t^{12} - 5400\,Bt^9x^2 - 8640\,Ct^8x^3 + 176\,Bt^{10} + 864\,Ct^9x - 3096\,t^{10}x - 1728\,A_4t^7x^3 + 17280\,Bt^7x^3 + 19440\,Ct^6x^4 - 1944\,Bt^8x - 1296\,A_3t^7x^2 - 4536\,Ct^7x^2 + 14580\,t^8x^2 + 9720\,A_4t^5x^4 - 29160\,Bt^5x^4 - 23328\,Ct^4x^5 + 96\,Ct^8 - 480\,t^9 - 864\,A_2t^7x + 7938\,Bt^6x^2 + 5832\,A_3t^5x^3 + 11448\,Ct^5x^3 - 32400\,t^6x^3 - 11664\,A_4t^3x^5 + 23328\,Bt^3x^5 + 11664\,Ct^2x^6 - 432\,A_1t^7 - 192\,Bt^7 - 864\,Ct^6x + 3888\,t^7x + 1944\,A_2t^5x^2 + 3456\,A_4t^4x^3 - 14310\,Bt^4x^3 - 13608\,Ct^3x^4 + 34020\,t^4x^4 - 5832\,A_4tx^6 - 5832\,Btx^6 - 1944\,A_1t^5x + 1296\,Bt^5x + 2592\,A_3t^4x^2 + 2592\,Ct^4x^2 - 10692\,t^5x^2 + 11664\,A_2t^3x^3 - 9720\,A_4t^2x^4 + 10206\,Bt^2x^4 - 17496\,A_3tx^5 + 5832\,Ctx^5 - 17496\,t^2x^5 - 5832\,A_0t^5 - 64\,Ct^5 + 216\,t^6 + 1728\,A_2t^4x + 23328\,A_1t^3x^2 - 2592\,Bt^3x^2 - 5832\,A_3t^2x^3 - 2808\,Ct^2x^3 + 9720\,t^3x^3 - 29160\,A_2tx^4 + 2916\,A_4x^5 - 1458\,Bx^5 + 8748\,x^6 + 864\,A_1t^4 + 80\,Bt^4 + 34992\,A_0t^3x + 288\,Ct^3x - 1080\,t^4x - 1944\,A_2t^2x^2 - 40824\,A_1tx^3 - 1728\,A_4tx^3 + 1404\,Btx^3 + 729\,Cx^4 + 972\,tx^4 + 1944\,A_1t^2x - 216\,Bt^2x - 52488\,A_0tx^2 - 1296\,A_3tx^2 - 216\,Ctx^2 + 1944\,t^2x^2 - 2916\,A_2x^3 + 5832\,A_0t^2 + 16\,Ct^2 - 864\,A_2tx - 5832\,A_1x^2 + 54\,Bx^2 + 324\,x^3 - 432\,A_1t - 8\,Bt - 8748\,A_0x + 288\,tx + 12\right)}{\left(-4\,t^7 + 8\,t^4 + 108\,tx^3 - 27\,(4\,t^3 - 1)x^2 + 36\,(t^5 - t^2)x - 4\,t\right)^{\frac{5}{2}}}$$

Let's extract the polynomial and see better what we are solving. **Note that we need to use the `operand` command to see which expression in the list is the polynomial, and extract it using an index.**

In [5]:
```
#extract the polynomial, other term in the expression doesn't matter
find_As_poly = find_As_exp.operands()[1]
show(QQ[A_0, A_1, A_2, A_3, A_4, B, C, t][x](find_As_poly))
```

$$\left(11664Ct^2 - 5832A_4t - 5832Bt + 8748\right)x^6 + \left(-23328Ct^4 - 11664A_4t^3 + 23328Bt^3 - 17496A_3t + 5832Ct - 17496t^2 + 2916A_4 - 1458B\right)x^5$$
$$+ \left(19440Ct^6 + 9720A_4t^5 - 29160Bt^5 - 13608Ct^3 + 34020t^4 - 9720A_4t^2 + 10206Bt^2 - 29160A_2t + 729C + 972t\right)x^4$$
$$+ \left(-8640Ct^8 - 1728A_4t^7 + 17280Bt^7 + 5832A_3t^5 + 11448Ct^5 - 32400t^6 + 3456A_4t^4 - 14310Bt^4 + 11664A_2t^3 - 5832A_3t^2 - 2808Ct^2 + 9720t^3 - 40824A_1t\right.$$
$$\left. - 1728A_4t + 1404Bt - 2916A_2 + 324\right)x^3$$
$$+ \left(2160Ct^{10} - 5400Bt^9 - 1296A_3t^7 - 4536Ct^7 + 14580t^8 + 7938Bt^6 + 1944A_2t^5 + 2592A_3t^4 + 2592Ct^4 - 10692t^5 + 23328A_1t^3 - 2592Bt^3 - 1944A_2t^2 - 52488A_0t\right.$$
$$\left. - 1296A_3t - 216Ct + 1944t^2 - 5832A_1 + 54B\right)x^2$$
$$+ \left(-288Ct^{12} + 864Bt^{11} + 864Ct^9 - 3096t^{10} - 1944Bt^8 - 864A_2t^7 - 864Ct^6 + 3888t^7 - 1944A_1t^5 + 1296Bt^5 + 1728A_2t^4 + 34992A_0t^3 + 288Ct^3 - 1080t^4 + 1944A_1t^2\right.$$
$$\left. - 216Bt^2 - 864A_2t - 8748A_0 + 288t\right)x + 16Ct^{14} - 56Bt^{13} - 64Ct^{11} + 252t^{12} + 176Bt^{10} + 96Ct^8 - 480t^9 - 432A_1t^7 - 192Bt^7 - 5832A_0t^5 - 64Ct^5 + 216t^6$$
$$+ 864A_1t^4 + 80Bt^4 + 5832A_0t^2 + 16Ct^2 - 432A_1t - 8Bt + 12$$

Note that we only need the numerator's $x^2, x^3 \cdots x^7$ coefficient be zero, so we extract them and use the `solve` command to find expressions for $A_0, \cdots A_4$.

In [6]:
```
# Remark, the coefficients property generates a list with
# [Coeff. of x^0, Coeff of x^1, ...]
coe_find_A = find_As_poly.coefficients(x, sparse=False)
# Coefficient of x^2, ... x^6 = 0
A_Coeff = solve(coe_find_A[2:7], A_0, A_1, A_2, A_3, A_4)
show(A_Coeff)
```

$$\left[\left[A_0 = \frac{1120\,Ct^{12} - 1680\,Bt^{11} - 3072\,Ct^9 + 4200\,t^{10} + 3744\,Bt^8 + 2874\,Ct^6 - 5472\,t^7 - 2556\,Bt^5 - 202\,Ct^3 + 1908\,t^4 + 6\,Bt^2 + 9\,C + 12\,t}{45360\,t^3}, A_1 = \right.\right.$$
$$-\frac{4480\,Ct^9 - 5600\,Bt^8 - 7664\,Ct^6 + 10080\,t^7 + 7048\,Bt^5 + 2428\,Ct^3 - 6564\,t^4 - 962\,Bt^2 + 27\,C + 876\,t}{15120\,t^2}, A_2$$
$$\left.\left. = \frac{160\,Ct^6 - 160\,Bt^5 - 136\,Ct^3 + 200\,t^4 + 82\,Bt^2 + 3\,C - 56\,t}{120\,t}, A_3 = -\frac{32\,Ct^5 - 24\,Bt^4 - 8\,Ct^2 + 24\,t^3 + 3\,Bt - 3}{12\,t^2}, A_4 = \frac{4\,Ct^2 - 2\,Bt + 3}{2\,t}\right]\right]$$

Note that the remaining two coefficient is

In [7]:
```
x1_coeff = coe_find_A[1]
x0_coeff = coe_find_A[0]
show(QQ[A_0, A_1, A_2, A_3, A_4, B, C][t](x0_coeff))
show(QQ[A_0, A_1, A_2, A_3, A_4, B, C][t](x1_coeff))
```

$$16Ct^{14} - 56Bt^{13} + 252t^{12} - 64Ct^{11} + 176Bt^{10} - 480t^9 + 96Ct^8 + \left(-432A_1 - 192B\right)t^7 + 216t^6 + \left(-5832A_0 - 64C\right)t^5 + \left(864A_1 + 80B\right)t^4 + \left(5832A_0 + 16C\right)t^2$$
$$+ \left(-432A_1 - 8B\right)t + 12$$

$$-288Ct^{12} + 864Bt^{11} - 3096t^{10} + 864Ct^9 - 1944Bt^8 + \left(-864A_2 + 3888\right)t^7 - 864Ct^6 + \left(-1944A_1 + 1296B\right)t^5 + \left(1728A_2 - 1080\right)t^4 + \left(34992A_0 + 288C\right)t^3$$
$$+ \left(1944A_1 - 216B\right)t^2 + \left(-864A_2 + 288\right)t - 8748A_0$$

For some reason they only depends on A_0, A_1, A_2, so strange!

Now we use our solved $A_0, A_1, \dots A_4$ to plug it in, and solve for both of the coefficient being zero.

In [8]:
```
one_solve = x1_coeff.subs(A_Coeff)
zero_solve = x0_coeff.subs(A_Coeff)
show(solve([one_solve, zero_solve], B, C))
```

$$\left[\left[B = \frac{4\,t^3 - 1}{t^4 - t}, C = \frac{2\,t}{t^3 - 1}\right]\right]$$

As consistent with the result presented.

Example 2

This part starts from Theorem 2.2 from Haessig.

First, note that the choice of our elliptic function in the last example have nothing in particular, except we need to inspect `operands` by hand and get the polynomial. That means the whole operation can be fixed into a function with only specifying `y`, the elliptic curve with square root of the normal form; and `index_fix` a helper variable so we can make sure we pick the numerator of the polynomial.

Let's do exactly that:

In [9]:
```
def find_Picard(y, index_fix=0):
    """
    Input elliptic curve y in y = sqrt(...), where
    ... is in normal form, display the coefficient B, C for
    the Picard-Fuchs Equation.
    """
    f = 1/y
    f_dot = f.diff(t)
    f_ddot = f_dot.diff(t)
    eta = f_ddot+B*f_dot+C*f
    eta = eta.factor()
    f_0 = A_0*(f^3)
    f_1 = A_1*x/(y^3)
    f_2 = A_2*x^2/(y^3)
    f_3 = A_3*x^3/(y^3)
    f_4 = A_4*x^4/(y^3)
    d_0 = f_0.diff(x)
    d_1 = f_1.diff(x)
    d_2 = f_2.diff(x)
    d_3 = f_3.diff(x)
    d_4 = f_4.diff(x)
    find_As_exp = eta+d_0+d_1+d_2+d_3+d_4
    find_As_exp = find_As_exp.factor()
    # index fix
    find_As_poly = find_As_exp.operands()[index_fix]
    coe_find_A = find_As_poly.coefficients(x, sparse=False)
    # Coefficient of x^2, ... x^6 = 0
    A_Coeff = solve(coe_find_A[2:7], A_0, A_1, A_2, A_3, A_4)
    x1_coeff = coe_find_A[1]
    x0_coeff = coe_find_A[0]
    one_solve = x1_coeff.subs(A_Coeff)
    zero_solve = x0_coeff.subs(A_Coeff)
    show(solve([one_solve, zero_solve], B, C))
    return None
```

Let's check our code is working well with the previous example:

In [10]:
```
Ex1 = sqrt(t*(x^3)+(1/4)*(1-4*t^3)*(x^2)+(1/3)* (t^2)*(t^3-1)*x-(1/27)*t*(t^3-1)^2)
find_Picard(Ex1, index_fix=1)
```

$$\left[\left[B=\frac{4\,t^3-1}{t^4-t},C=\frac{2\,t}{t^3-1}\right]\right]$$

Good. Now straight onto the two example, let's do $p=3$ first.

$$Y^2=X^3+X^2-t$$

In [11]:
```
Ex2=sqrt(x^3+x^2-t)
find_Picard(Ex2)
```

$$\left[\left[B=\frac{2\,(27\,t-2)}{27\,t^2-4\,t},C=\frac{15}{4\,(27\,t^2-4\,t)}\right]\right]$$

Now $p=2$! For some reason the thing

$$Y^2+XY=X^3+tX$$

, is not in normal form, but this can always be done:

> **Theorem.** Given an elliptic curve of the form
>
> $$y^2+a_1xy+a_3y=x^3+a_2x^2+a_4x+a_6$$
>
> Then the transformation
>
> $$y\longrightarrow y-\frac{a_1x+a_3}{2},x\longrightarrow x-\frac{A}{3}$$
>
> ,

where A is the coefficient of x^2 after the first transformation,

> performed in this order, always gives an elliptic curve of standard for $y^2=x^3+ax+b$

(Theorem taken from Here.)

So our required transformation is $y\to y-\frac{x}{2}$ first:

In [12]:
```
y = var("y")
eq = y^2+x*y == x^3+t*x
eq = eq.subs(y == y-x/2)
show(eq.expand())
```

$$-\frac{1}{4}\,x^2+y^2=x^3+tx$$

So the equation under basis $(x,y)=\{(1,0),(-1/2,1)\}$ is

$$y^2=x^3+\frac{1}{4}x^2+tx$$

Note that this is not in normal form, but we can still try to find its Picard-Fuchs Equation:

In [13]:
```
Ex3 = sqrt(x^3+(1/4)*x^2+t*x)
find_Picard(Ex3)
```

$$\left[\left[B = \frac{128\,t - 1}{64\,t^2 - t}, C = \frac{12}{64\,t^2 - t}\right]\right]$$

Does this change if we put the above equation into its actual normal form? To do this, by the above theorem we do $x \to x - \frac{1}{12}$:

In [18]:
```
E = y^2 == x^3+(1/4)*x^2+t*x
F = E.subs(x == x-1/12)
show(F.expand())
```

$$y^2 = x^3 + tx - \frac{1}{12}\,t - \frac{1}{48}\,x + \frac{1}{864}$$

So the actual normal form is:

$$y^2 = x^3 + \left(t - \frac{1}{48}\right)x - \frac{1}{12}t + \frac{1}{864}$$

We can find its Picard-Fuchs Equation again:

In [15]:
```
Ex4 = sqrt(x^3+(t-1/48)*x-(1/12)*t+1/864)
find_Picard(Ex4, index_fix=1)
```

$$\left[\left[B = \frac{128\,t - 1}{64\,t^2 - t}, C = \frac{12}{64\,t^2 - t}\right]\right]$$

It doesn't change! So at least we are partly sure this is okay...